The CAPM

Result from last time

When markets are in equilibrium
$\Updownarrow$
Demand = Supply

the ORP is the Market Portfolio

ORP: optimal risky portfolio = tangency T

Market Portfolio: each asset's weight is the value-weight = $\dfrac{\text{Market Cap}}{\text{Total Market Cap}}$

$\Rightarrow$ From now on, we'll work with the Market, M.

The market portfolio has the highest Sharpe's Ratio:

$$\boxed{SR(M) = \underline{\underline{\text{Highest}}}}$$

Suppose we build a portfolio P:

Assets:	Market, M	Asset i
Weights:	$1-a$	a

Need $SR(P) = ?$

$$SR(P) = \frac{E(r_P) - r_F}{\sigma(r_P)}$$

$$E(r_P) = E(r_M) \times (1-a) + E(r_i) \times a$$

$$\sigma^2(r_P) = \sigma^2(r_M)(1-a)^2 + \sigma^2(r_i)\,a^2 + 2a(1-a)\,\mathrm{cov}(r_M, r_i)$$

$$SR(P) = \frac{\boxed{E(r_M)}(1-a) + \boxed{E(r_i)}a - \boxed{r_F}}{\sqrt{\boxed{\sigma_M^2}(1-a)^2 + \boxed{\sigma_i^2}a^2 + 2a(1-a)\boxed{cov(r_M, r_i)}}}$$

We want to find a that achieves: $\boxed{\max SR(P)}$

We already know the answer: $\boxed{a = 0}$

Calculus:

Take a derivative of SR(P) wrt $\underline{a}$:

$$SR(P)' = 0$$

After calculating the derivative $SR(P)'$ and pluging in $a=0$ we obtain:

$$\frac{E(r_i)-r_F}{E(r_M)-r_F} = \underbrace{\frac{\mathrm{cov}(r_i, r_M)}{\sigma^2(r_M)}}_{\beta_i}$$

$$\frac{E(r_i)-r_F}{E(r_M)-r_F} = \beta_i$$

$$\boxed{E(r_i)-r_F = \beta_i\,(E(r_M)-r_F)}$$

the CAPM